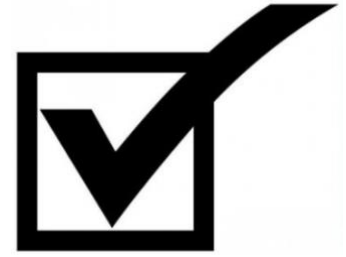




**BUSINESS
PROFESSIONALS**
of **AMERICA**
Giving Purpose to Potential



MANAGERIAL ACCOUNTING (135)

STATE 2022

CONCEPT KNOWLEDGE:

Multiple Choice (15 @ 2 points each) _____ (30 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (87 points)

Problem 1: Production Cost Report _____ (68 points)

Problem 2: Budgeting Scenarios _____ (45 points)

TOTAL POINTS _____ ***(230 points)***

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

MULTIPLE CHOICE (10 @ 2 points each = 20 points)

- | | | |
|------|-------|-------|
| 1. A | 6. C | 11. B |
| 2. C | 7. D | 12. B |
| 3. D | 8. B | 13. D |
| 4. C | 9. B | 14. D |
| 5. A | 10. C | 15. A |

Short Answer (Each answer worth 3 points; total 87 Points)

1. O'Brien Corporation, a merchandising company, reported the following results for October:

Sales	\$413,000
Cost of goods sold (all variable)	\$173,100
Total variable selling expense	\$23,000
Total fixed selling expense	\$19,300
Total variable administrative expense	\$17,400
Total fixed administrative expense	\$31,000

What is the contribution margin for October?

Contribution Margin for October	\$199,500
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2. Saunders Corporation produces and sells a single product. Data concerning that product appear below:

	Per Unit	Percent of Sales
Selling price	\$ 140	100 %
Variable expenses	84	60 %
Contribution margin	<u>\$ 56</u>	<u>40 %</u>

The company is currently selling 5,300 units per month. Fixed expenses are \$185,000 per month. The marketing manager believes that a \$5,300 increase in the monthly advertising budget would result in a 200 unit increase in monthly sales. What should be the overall effect on the company's monthly net operating income of this change?

Increase of \$5,900

3. McNew Inc. produces and sells a single product. The selling price of the product is \$200.00 per unit and its variable cost is \$40.00 per unit. The fixed expense is \$407,820 per month. Round the units to the correct whole number and the dollars to the nearest cent.

Break-even point in units	2,549
Break-even point in sales dollars	\$509,775
How many units does McNew need to sell to earn \$250,000 in profit?	4,112

4. Bobb Corporation's contribution format income statement for July appears below:

Sales	\$ 273,600
Variable expenses	<u>123,120</u>
Contribution margin	150,480
Fixed expenses	<u>49,660</u>
Net operating income	<u>\$ 100,820</u>

What is the degree of operating leverage?

Degree of Operating Leverage	1.49
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5. Spangler Inc. has an operating leverage of 4.8. If the company's sales increase by 13%, its net operating income should increase by what percentage?

Percentage increase in operating income	62.4%
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6. The Refining Department of Weber, Inc. had 63,000 tons of sugar to account for in December. Of the 63,000 tons, 45,000 tons were completed and transferred to the Boiling Department, and the remaining 18,000 tons were 60% complete with respect to conversion costs. The materials required for production are added at the beginning of the process. Conversion costs are added equally throughout the refining process. The weighted-average method is used.

What are the total equivalent units of production for direct materials?	63,000
What are the total equivalent units of production for conversion costs?	55,800

7. Hendrickson Corporation has provided the following information:

	Cost per Unit	Cost per Period
Direct materials	\$6.95	
Direct labor	\$4.00	
Variable manufacturing overhead	\$1.45	
Fixed manufacturing overhead		\$109,800
Sales commissions	\$1.05	
Variable administrative expense	\$0.75	
Fixed selling and administrative expense		\$40,050

What is the incremental manufacturing cost that the company will incur if it increases production from 9000 to 9001 units?

Incremental Manufacturing Cost	\$12.40
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8. Hewes Inc. produces and sells two products. Data concerning those products for the most recent month appear below:

	Product DE	Product AH
Sales	\$ 51,000	\$ 56,000
Variable expenses	\$ 13,900	\$ 26,760

The fixed expenses of the entire company were \$39,050.

What is the break-even point in sales dollars for the entire company?	\$62,983.87
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What is the break-even point in sales dollars for each product?

Product DE	30,020.35
Product AH	32,963.52

9. Martin Co.'s phone bill consists of both fixed and variable costs. Refer to the four-month data below and apply the high-low method to answer the question.

	Minutes	Total Bill
January	460	\$3,025
February	240	\$2,675
March	210	\$2,500
April	320	\$2,840

What is the variable cost per minute?	\$2.10
What is the fixed portion of the total cost?	\$2,059
If 390 minutes are used in May, how much will the total bill be?	\$2,878

10. Schultz Company has the following data for the Assembly Department for August:

Units in process at the beginning of August	800
Units started in August	2,000
Units completed and transferred	2,100
Units in process at end of August	700

Conversion costs are added evenly throughout the process. The company uses the weighted-average method. Compute the equivalent units of production for direct materials and conversion costs for each independent scenario:

	Direct Materials	Conversion Costs
Units in process at the end of August are 20% complete; materials are added at the beginning of the process.	2,800	2,240
Units in process at the end of August are 80% complete; materials are added at the beginning of the process.	2,800	2,660
Units in process at the end of August are 30% complete; materials are added at the end of the process.	2,100	2,310
Units in process at the end of August are 75% complete; materials are added at the halfway point.	2,800	2,625

11. Nelson & Lott Co. incurred the following costs while producing 500 units: direct materials, \$10 per unit; direct labor, \$25 per unit; variable manufacturing overhead, \$15 per unit; total fixed overhead costs, \$10,000; variable selling and administrative costs, \$5 per unit; total fixed selling and administrative costs, \$7,500. There are no beginning inventories.

What is the unit product cost using variable costing?	\$50
What is the unit product cost using absorption costing?	\$70
What is the operating income using absorption costing if 450 units are sold for \$100 each?	\$3,750
What is the operating income using variable costing if 450 units are sold for \$100 each?	\$2,750

Problem 1 (2 points for each unshaded square; total 68 Points)

Robsplach Co. prepares and packages fertilizer for commercial landscaping business. Robsplach has two departments: Mixing and Packaging. Direct materials are added at the beginning of the mixing process and at the end of the packaging process. Conversion costs are added evenly throughout each process. Data from the month of May for the Mixing Department are as follows:

Gallons		Costs	
Beginning Work-in-Process Inventory	0 gallons	Beginning Work-in-Process Inventory	\$0
Started in production	8,500 gallons	Costs added during May:	
Completed and transferred out to Packaging in May	6,000 gallons	Direct materials	\$5,525
Ending Work-in-Process Inventory (10% of the way through the blending process)	2,500 gallons	Direct labor	1,500
		Manufacturing overhead allocated	<u>2,125</u>
		Total costs added during May	<u>\$ 9,150</u>

Prepare the Production Cost Report for the Mixing Department on the next page by filling in all the unshaded box.

Robsplach Co.				
Production Cost Report – Mixing Dept				
Month Ended May 31, 2022				
		Equivalent Units		
Units	Whole Units	Direct Materials	Conversion Costs	
Units to account for:				
Beg WIP	0			
Started into Production	8,500			
Total units to account for	8,500			
Units accounted for				
Completed & transferred out	6,000	6,000	6,000	
End WIP	2,500	2,500	250	
Total units accounted for	8,500	8,500	6,250	
Costs		Direct Materials	Conversion Costs	Total Costs
Costs to account for				
Beg WIP		\$0	\$0	\$0
Costs added during period		5,525	3,625	9,150
Total costs to account for		\$5,525	\$3,625	\$9,150
Divided by Total EUP		8,500	6,250	
Cost per EUP		\$0.65	\$0.58	
Costs accounted for				
Completed & transferred out		\$3,900	\$3,480	\$7,380
End WIP		1,625	145	1,770
Total costs accounted for		\$5,525	\$3,625	\$9,150

Problem 2 (Each answer worth 5 points; total 45 Points)

Each of the following budget scenarios should be answered independently.

A) Thomas Company makes a product that is very popular as a Christmas gift. Thus, peak sales occur in December of each year, as shown in the sales budget for the fourth quarter given below:

	October	November	December	Total
Budgeted Sales (all on account)	\$200,000	\$300,000	\$500,000	\$1,000,000

From past experience, the company has learned that 20% of a month's sales are collected in the month of sale, 70% are collected in the month following sale, and the remaining 10% are collected in the second month following sale. Bad debts are negligible and can be ignored.

How much cash is expected to be collected in December?	\$330,000
What will be the balance in Accounts Receivable at the end of December?	\$430,000

B) Swenson Company has budgeted sales of its popular moose harness for the next four months as follows:

	Unit Sales
January	50,000
February	75,000
March	90,000
April	80,000

The company is now in the process of preparing a production budget for the first quarter. Past experience has shown that end-of-month inventory levels must equal 10% of the following month's unit sales. The inventory at the end of December was 5,000 units.

How many units should the company plan to produce in January?	52,500
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C) Three grams of rose oil are required for each bottle of PBA, a very popular perfume made by a small company in Ohio. The cost of rose oil is \$1.50 per gram. Budgeted production of PBA is given below for the last quarter of Year 2 and the first quarter of Year 3.

	Year 2			Year 3
	October	November	December	January
Budgeted Production (in bottles)	60,000	90,000	150,000	70,000

The inventory of rose oil at the end of the month must be equal to 20% of the following month's production needs. Approximately 12,000 grams of rose oil will be on hand at the start of October.

How many grams of rose oil should be purchased on October?	66,000
How much will the rose oil cost for the October purchases?	\$99,000
The company pays for 60% of the purchases in the month of purchase, and 40% in the month following purchase. Accounts Payable at the end of September is \$90,000. How much will the cash payments be for October?	\$149,400
What will be the balance in Accounts Payable at the end of October?	\$39,600

D) Lowe Inc. expects to produce 8,000 units in the next quarter. Each unit requires 0.35 direct labor hours, and each employee is paid \$12.00 per hour.

How many hours of direct labor should Lowe budget for?	2,800
How much direct labor cost should Lowe budget for?	\$33,600